LECTURE NOTES

Lecture 13: Exam preparation

AI504 - Knowledge Representation

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Contents

1 Latin squares

types:

- Set of cells $G = x_1, ... x_9$
- Set of rows R
- set of columns C
- Set of characters A (alphabet)

One would need 3 functions

```
r :: Eq => G -> R
c :: Eq => G -> C
a :: Eq => G -> A
```

A model $\mathcal{M}$ in this signature would be called a configuration.

$$\forall x \in R, \forall y \in A, \exists z \in G : r(z) = x \wedge a(z) = y$$

$$\forall x \in C, \forall y \in A, \exists z \in G : c(z) = x \wedge a(z) = y$$

How do we then express uniqueness?

$$\forall z_1, z_2 \in G : ((r(z_1) = r(z_2) \vee c(z_1) = c(z_2)) \wedge a(z_1) = a(z_2) \Rightarrow z_1 = z_2)$$

2 First-order logic

Signatures: in full generality these can consist of a set of types plus a set of typed function adn relation symbols

Example: Latin Squares

4 types R, C, A, G

plus function symbols

```
r :: Eq => G -> R
c :: Eq => G -> C
a :: Eq => G -> A
```